

Operating Notes for the Ridgeline Weather Station Network

These notes cover the day-to-day operation of the eight automatic weather stations that make up the Ridgeline network. They are written for the volunteers who take a turn on the maintenance rota, and they assume no background in meteorology beyond a willingness to read a dial carefully and write down what it says. Where a procedure depends on a particular instrument the model is named; where it does not, treat the description as general advice.

The network exists because the regional forecast grid is too coarse to say anything useful about the valley. Cold air pools along the river on clear nights, and the difference between the ridge and the valley floor can reach nine degrees before dawn. A single station at either end would miss that gradient entirely, so the design distributes sensors along the slope and accepts the extra maintenance burden that comes with eight sites instead of one.

Newcomers are often surprised at how much of the work is bookkeeping rather than electronics. Expect to spend more time reconciling logs than holding a screwdriver. The instruments themselves are reliable to the point of being dull, and the failures that actually cost us data have almost always been clerical: a site visit recorded against the wrong station, a battery swap that nobody noted, a calibration applied twice because two people each assumed the other had skipped it.

Each station reports every ten minutes over a low-power radio link to the base receiver in the old pump house. The link is deliberately simple and deliberately slow. It carries temperature, humidity, wind speed, wind direction and rainfall, and nothing else. Adding a channel means reflashing every node in the field, which is a full weekend of work, so proposals to measure soil moisture or solar irradiance have so far been declined on grounds of effort rather than merit.

Data lands in a single append-only file per station per month. Nothing is ever overwritten. If a reading is wrong we annotate it in a separate corrections file and leave the original in place, because the raw series is the only durable record of what the instrument actually reported. This convention has saved us twice, once after a firmware bug and once after a well-meaning volunteer recalibrated a thermometer without writing down the offset.

The maintenance rota runs on a four-week cycle, with each volunteer taking two stations. Pair the high sites with the low ones so that nobody spends every visit climbing. The rota is posted in the pump house and mirrored in the shared calendar, and the two are expected to agree; when they do not, the printed copy in the pump house is authoritative, because it is the one people actually read while standing in the rain.

A routine visit has four parts: inspect, clean, verify, record. Inspect means looking at the mast, the guy lines and the enclosure seals before touching anything. Clean means the radiation shield and the rain gauge funnel, in that order, using water and a soft brush and never a solvent. Verify means comparing the live reading against a hand-held reference. Record means writing the visit into the site book before you leave the site.

Wind cups deserve particular attention because they fail gradually rather than suddenly. A bearing that has begun to dry out will still turn, and the station will still report a plausible wind speed, but the low-end response degrades early and calm conditions start reading as dead calm. Spin the cups by hand and listen; a healthy assembly coasts for several seconds with no audible roughness.

The rain gauge is the instrument most likely to disagree with its neighbours, and almost always for mechanical reasons. Check that the funnel is level, that the tipping bucket swings freely through its full travel, and that no spider has decided the collector is a good place to live. A gauge reading low by a third is far more often blocked than broken.

Battery condition is checked at every visit and logged as a voltage, not as a verdict. Do not write good or weak in the site book. Write the number. A voltage series tells you when a cell is drifting toward failure months before a verdict would, and the whole point of the log is to let the next volunteer see a trend rather than an opinion.

Anything you cannot fix in twenty minutes should be photographed and left alone. The base receiver flags a missing station within an hour, so a site that goes quiet is not a silent failure, and a half-finished repair is considerably worse than an honest outage. Note what you saw, note what you tried, and hand it on.

Calibration is the part of the work that most rewards patience. Every sensor drifts, most of them slowly and predictably, and the purpose of a calibration visit is not to make a reading correct today but to characterise how it has moved since the last visit. That means the reference instrument matters more than the station instrument, and the reference lives in a padded case and never leaves the pump house except on a calibration round.

Temperature is compared inside the radiation shield, not beside it, and the comparison is taken after the reference has had ten full minutes to equilibrate. Rushing this is the single most common source of spurious offsets in our records. If the two readings differ by more than a quarter of a degree after equilibration, note it and move on; do not adjust anything in the field.

It is worth saying plainly that a calibration round takes a full day for the whole network and should not be attempted in poor weather. Wet hands, a cold reference instrument and a hurried comparison will produce numbers that look like data and behave like noise. If the forecast turns while you are out, abandon the round and record how far you got; a partial round with honest notes is genuinely useful, whereas a complete round done badly is worse than none at all.

Humidity sensors are replaced rather than calibrated. The capacitive elements we use age in a way that is not usefully correctable, and a replacement costs less than an afternoon of comparison work. Replace them on a two-year cycle regardless of apparent health, and record the serial number of the new element in the site book so that a suspicious series can later be tied to a specific part.

Wind direction is checked against a surveyed bearing marked on the mast base. The mark is easy to miss under leaf litter, so clear the base before you start. An error here is insidious because the data remain entirely credible: a vane offset by thirty degrees produces a wind rose that is merely wrong rather than obviously broken.

The base receiver is a small computer in a sealed box on the pump house wall. It has no screen and no keyboard by design. Everything it does is visible in its log file, and everything you might want to change lives in a single configuration file beside that log. If you find yourself wanting to attach a monitor, what you actually want is to read the log from the shared drive.

Reception problems are almost never the radio. Before touching the receiver, check the obvious physical causes in order: a fallen branch across the line of sight, a bird nest on the antenna, water in the connector, and a corroded ground strap. Each of these has taken a station off the network at least once, and none of them look like a radio fault in the log.

The log records a heartbeat from every station and a checksum for every packet. A station that reports intermittently with clean checksums has a power problem; a station that reports constantly with bad checksums has a link problem. Learning to read that distinction quickly is the most valued skill a volunteer can pick up, and it takes about three visits to acquire.

Time synchronisation matters more than it appears to. The stations timestamp their own readings, and a node whose clock has drifted will place a rainfall burst in the wrong ten-minute bucket. The receiver corrects small drifts automatically and complains about large ones. A complaint in the log about a clock offset is worth acting on the same week.

Backups are pulled nightly to the shared drive and weekly to a disk that lives off site. The off-site copy is rotated by hand, which means it is only as good as the person whose turn it is. Sign the rotation sheet when you swap the disk. An unsigned sheet has, in practice, always meant the swap did not happen.

The configuration file is under version control and every change wants a one-line reason. Not what you changed, which is visible in the diff, but why. Six months later the diff is still legible and the reason never is unless somebody wrote it down at the time.

Reporting closes the loop. The network exists to answer questions about the valley, and a reading that nobody ever looks at is indistinguishable from a reading that was never taken. The monthly summary is therefore not an administrative chore but the actual output of all the maintenance work described above.

The summary is short on purpose: one page, four charts and a paragraph of commentary. Anyone who wants more can read the raw series. What the one page must do is make an unusual month obvious to somebody who is not looking for anything in particular, which rules out dense tables and rules in large, plain charts with honest axes.

Gaps are reported, never quietly interpolated. If a station was down for three days, the chart shows a gap three days wide and the commentary says why. We have been asked more than once to smooth these over for presentation purposes and have declined every time, because a network that hides its outages cannot be used to argue about anything.

Finally, keep the site books. They are paper, they are duplicated badly, and they are the only place where the reasoning behind a decade of small decisions is written down. When a station is decommissioned its book goes into the archive box, not the recycling. Somebody will want it.